

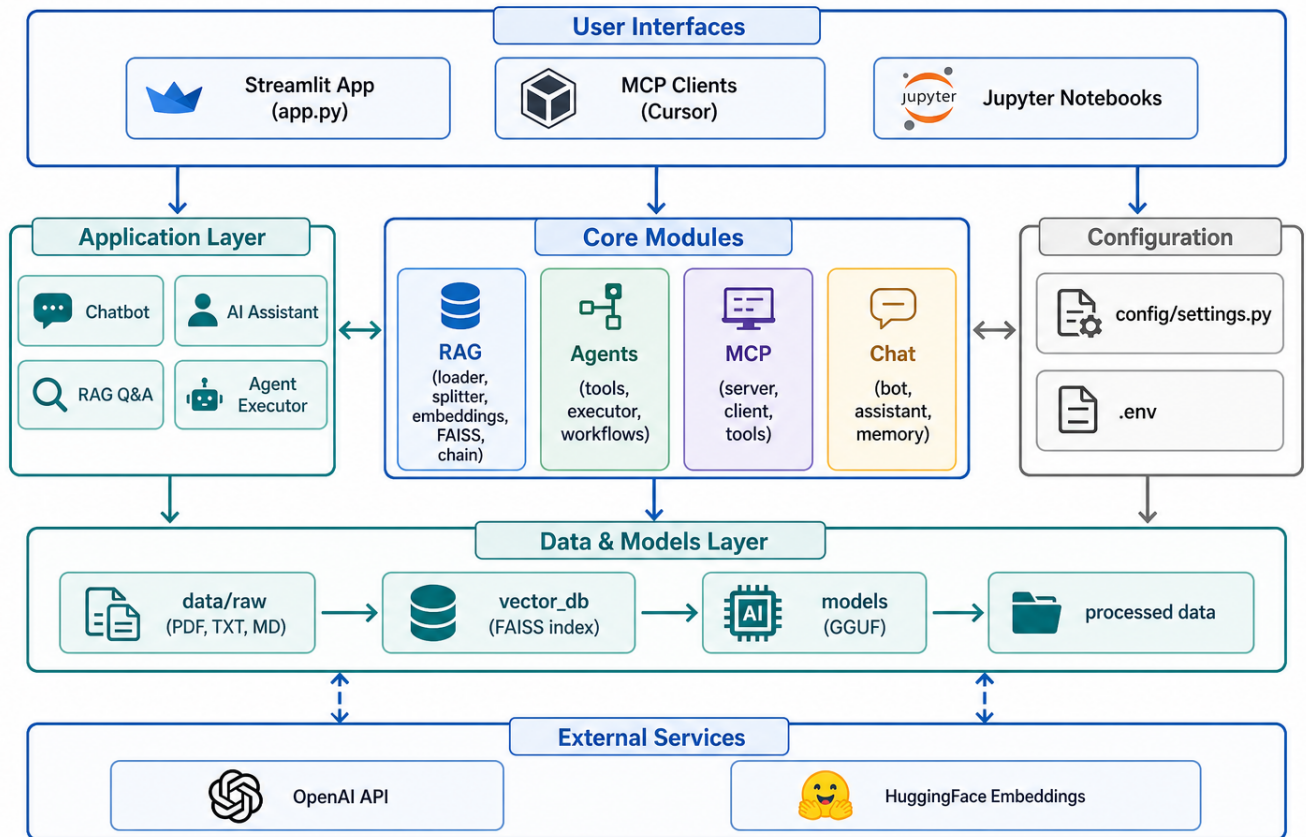
# Flow Diagrams

RAG, Agent, MCP, and Application Modes

Project: AI\_ML\_GENAI

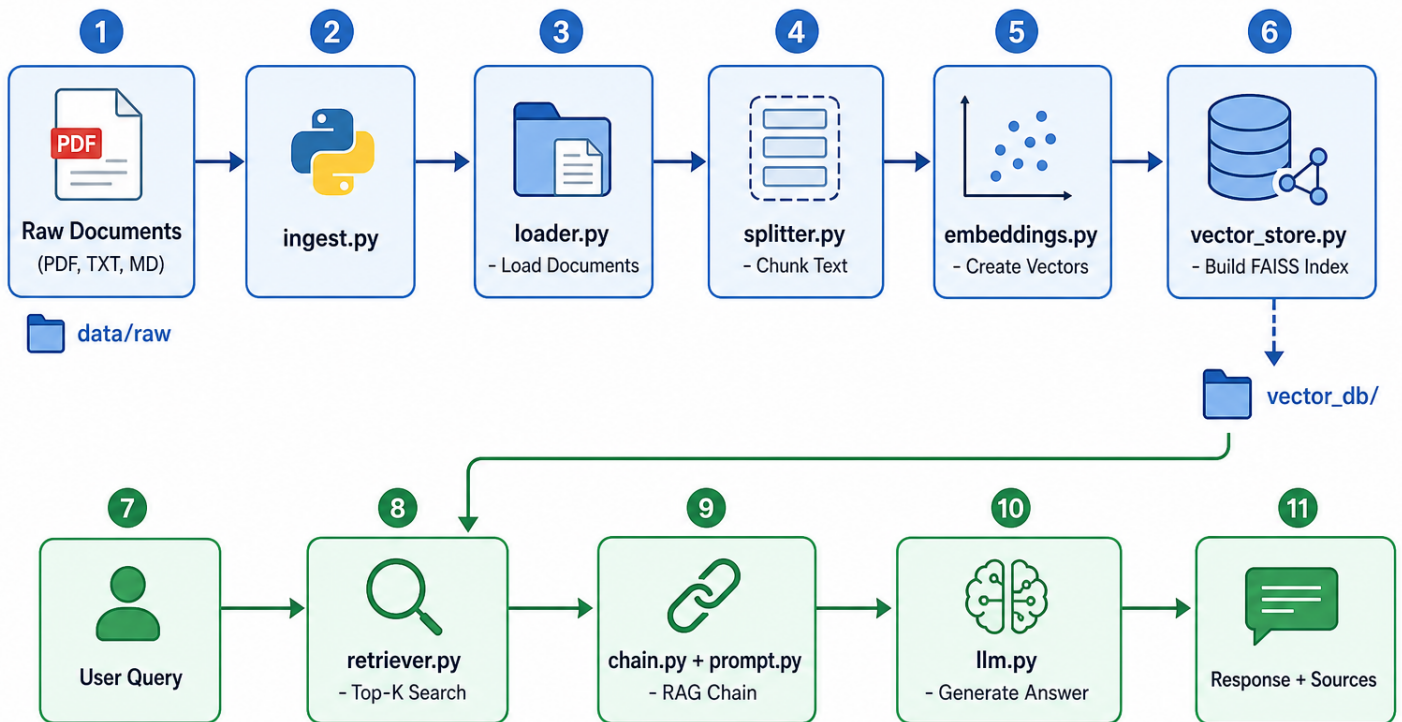
## System Architecture Overview

## AI ML GenAI - System Architecture



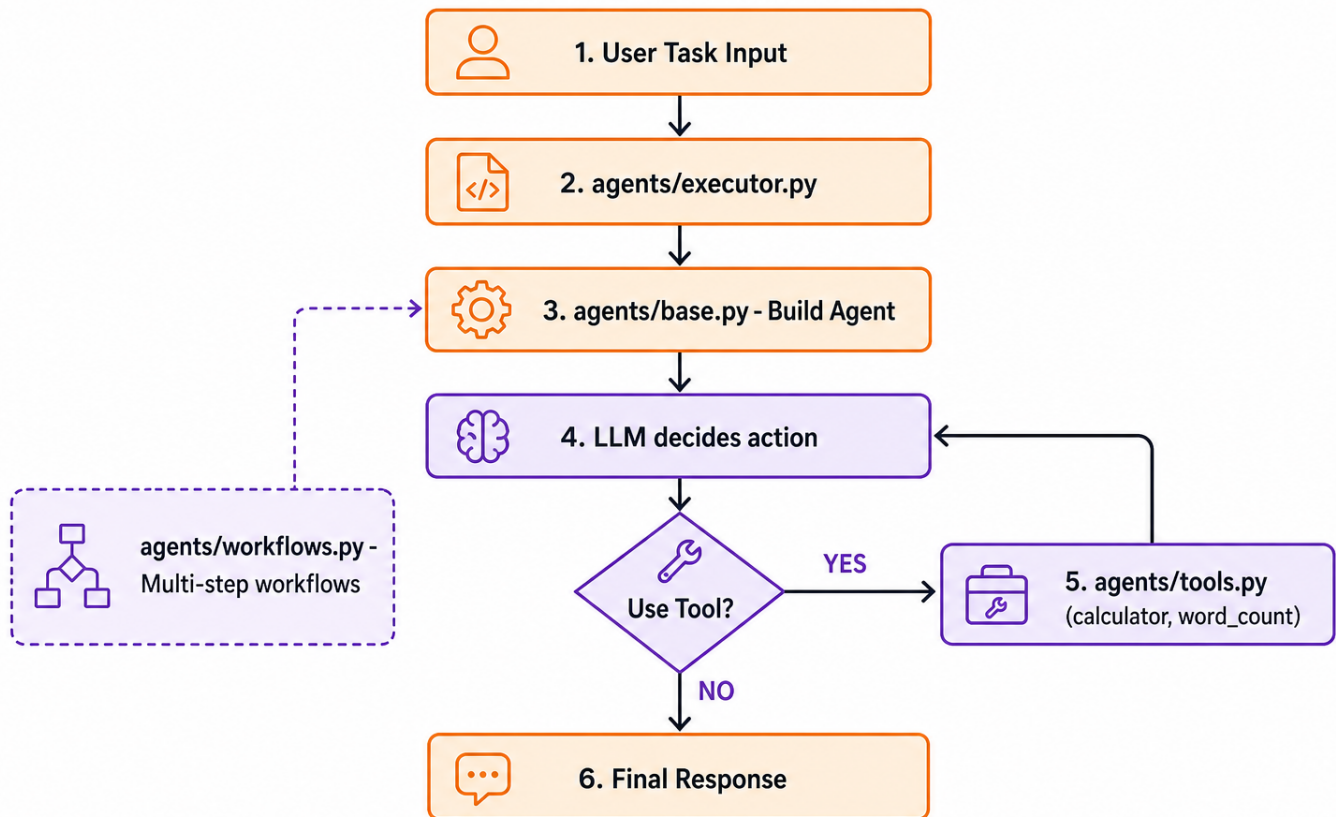
## RAG Pipeline Flow

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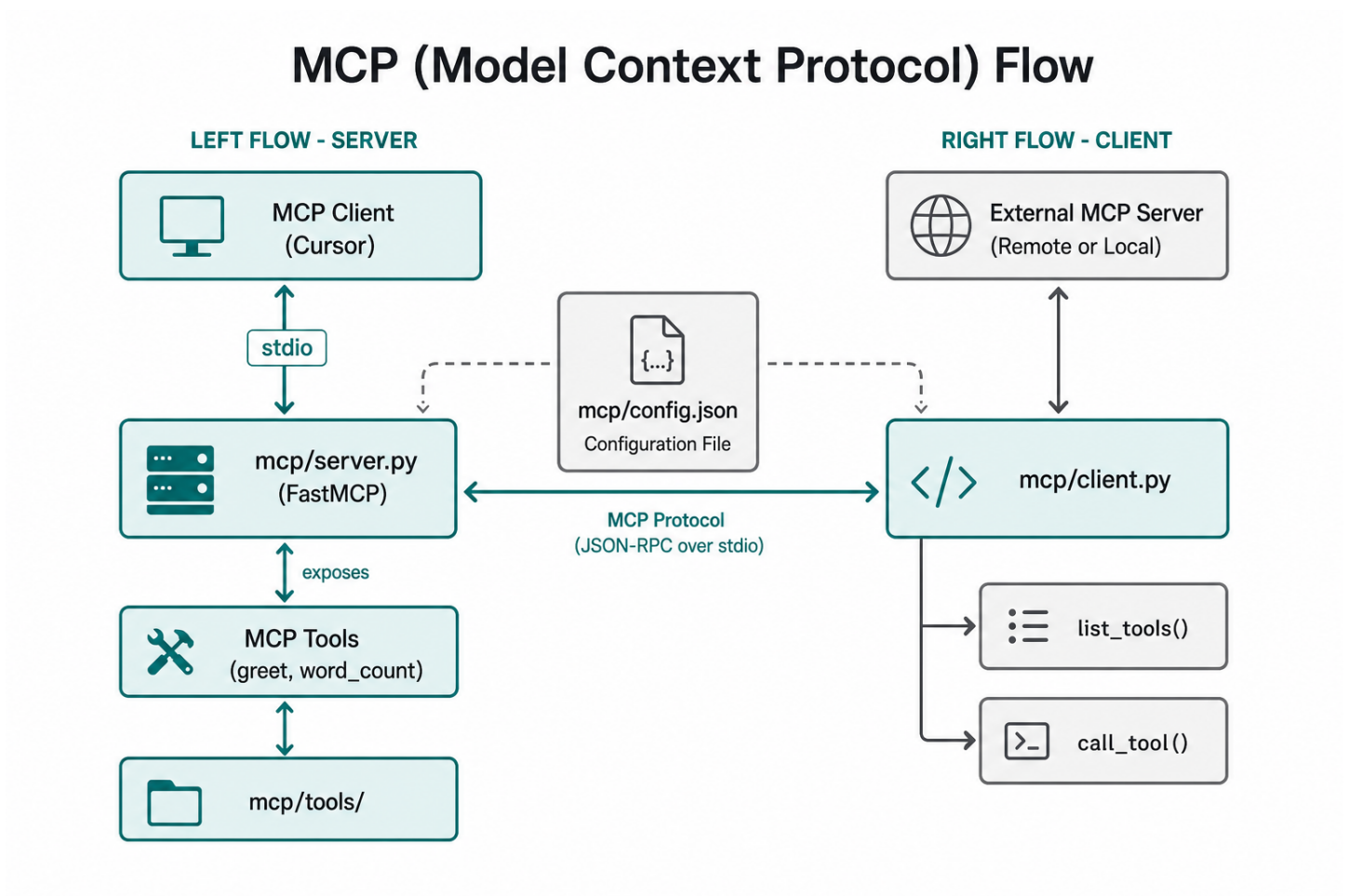


## AI Agent Flow

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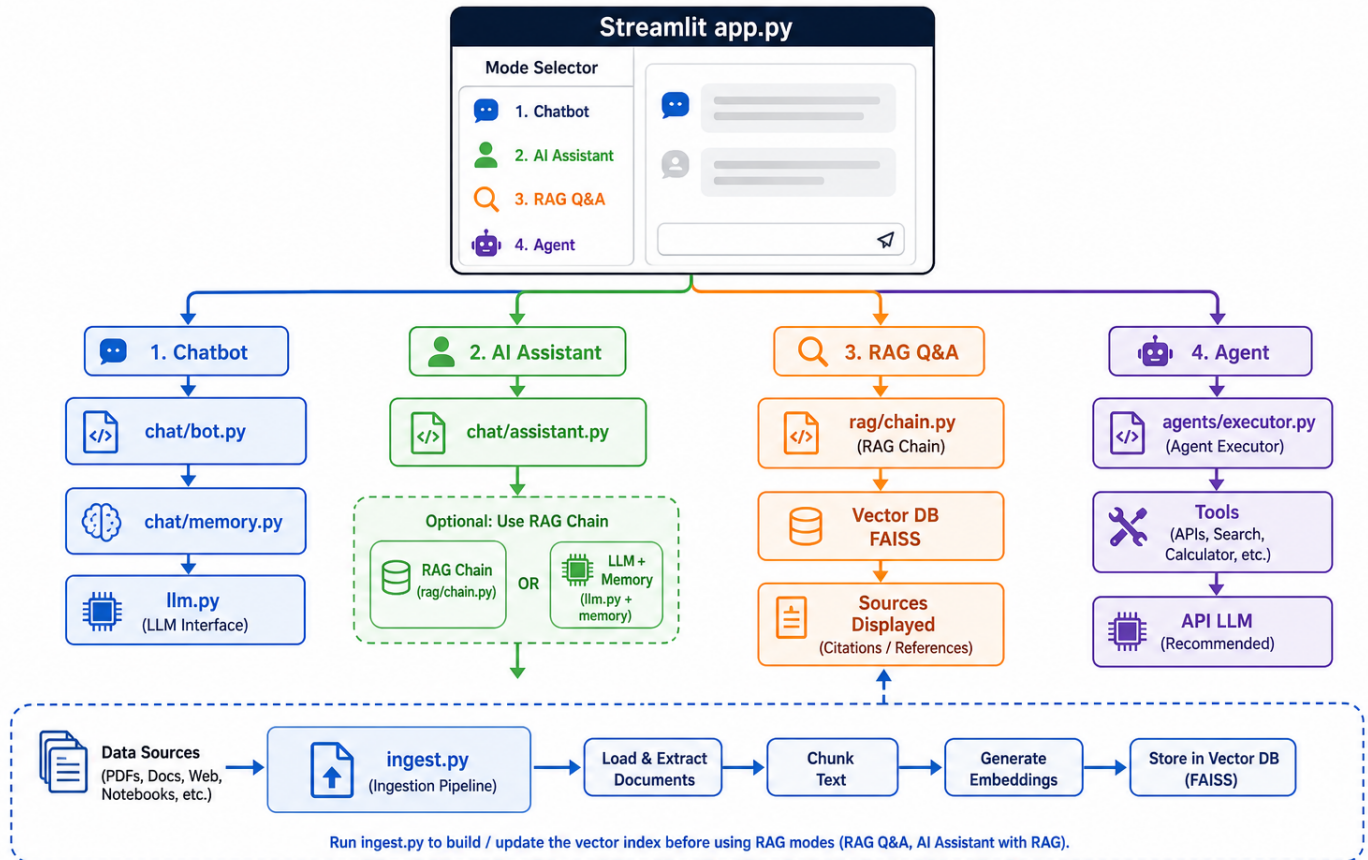
## MCP (Model Context Protocol) Flow





## Streamlit Application Modes

## Application Modes - app.py Streamlit UI



# Flow Diagrams

Step-by-step flows for each major capability. Each section includes a PNG diagram and a Mermaid flowchart (for editors that support it).

## 1. Overall application flow

flowchart TD

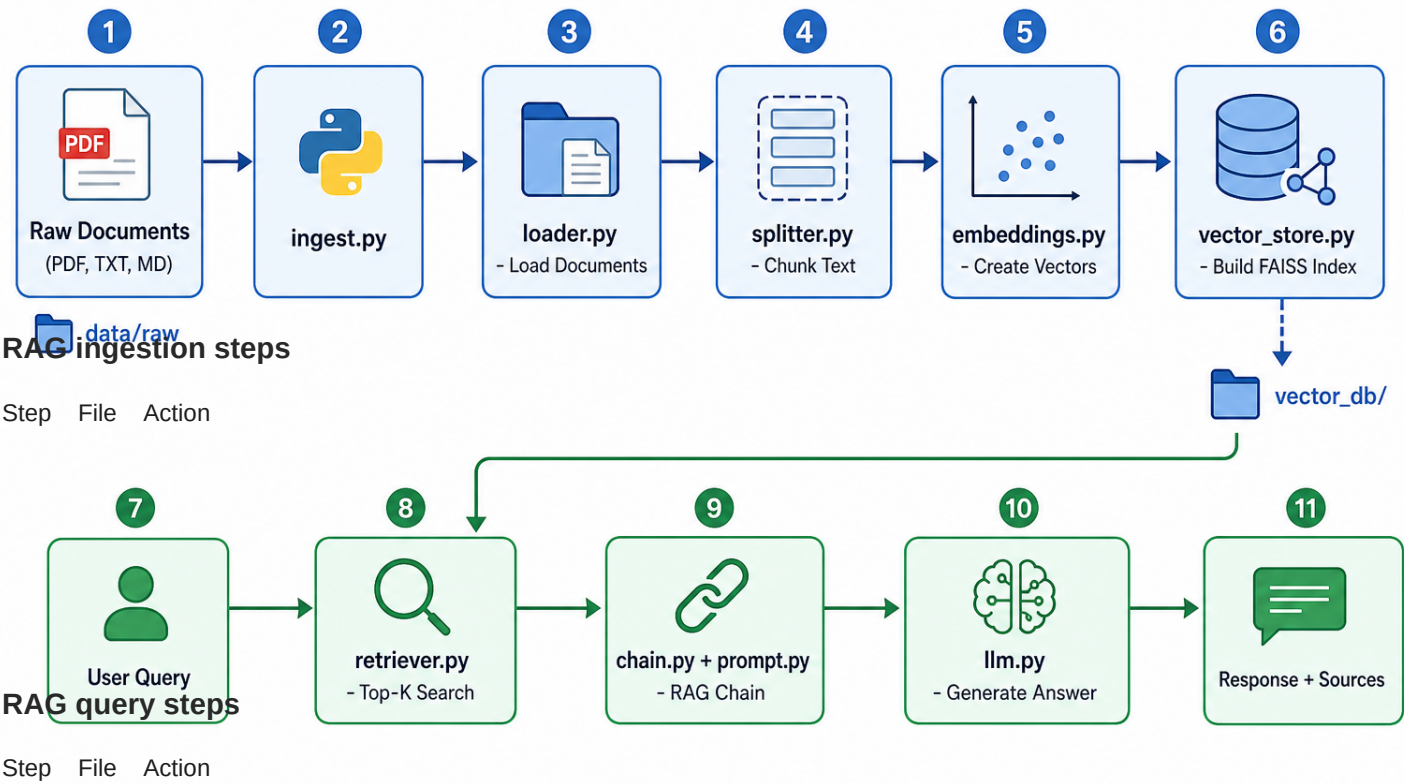
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## 2. RAG pipeline flow

RAG pipeline flow

flowchart LR

RAG Pipeline Flow



3. Agent flow



Agent flow

flowchart TD

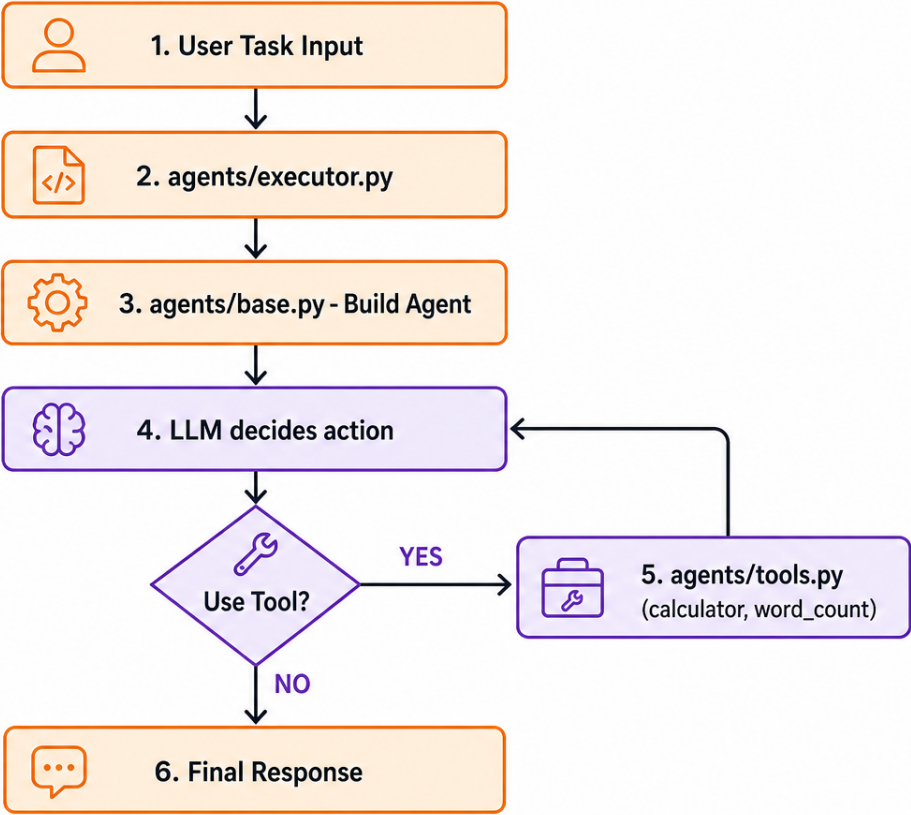
Agent loop

1. User submits task via Agent mode



4. MCP flow

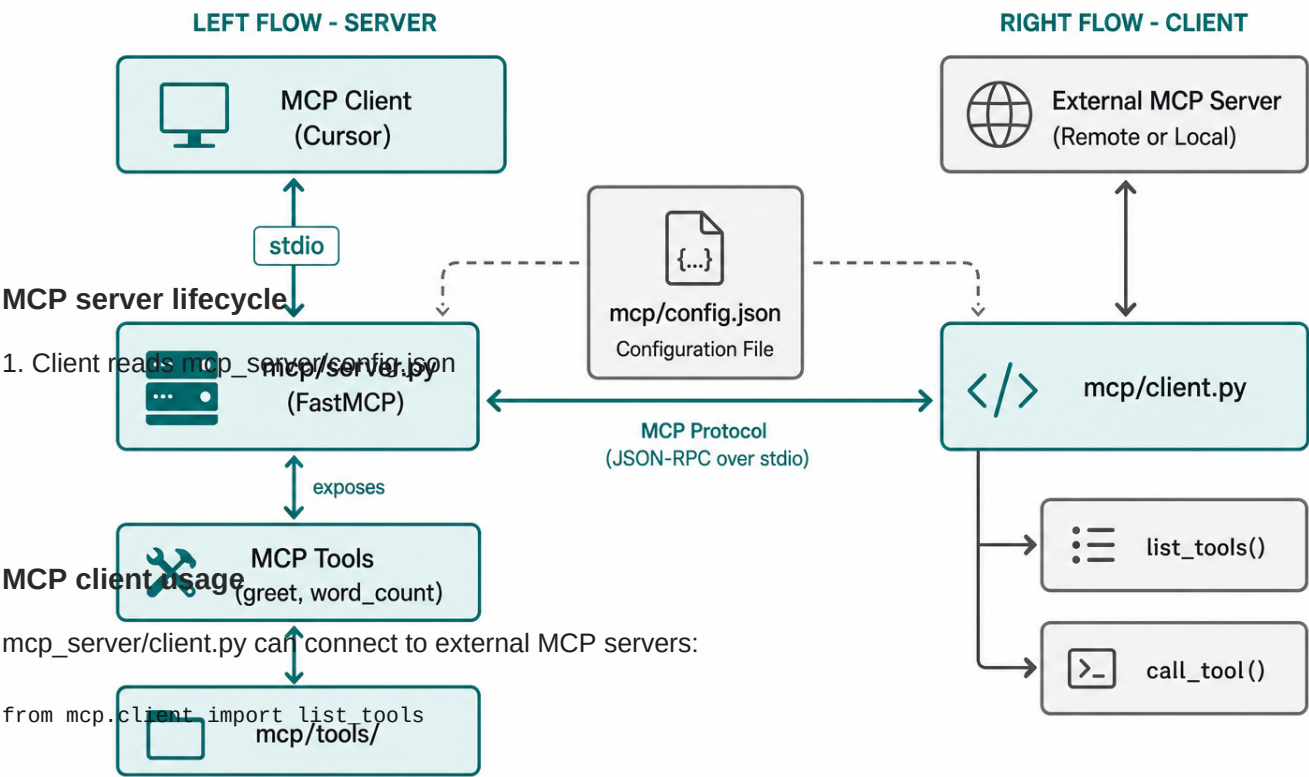
AI Agent Flow



MCP flow

flowchart LR

MCP (Model Context Protocol) Flow



5. Chatbot & assistant flow

flowchart TD

6. Document ingestion decision flow

flowchart TD

## 7. LLM selection flow

flowchart TD

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